

BRIER ISLAND, NS, Canada

WMO: 719880

Lat: **44.286N**Lon: **66.343W**Elev: **52**StdP: **14.67**Time Zone: **-4.00 (NAA)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.1	13.8	5.1	7.2	13.7	7.4	8.1	16.0	47.9	26.4	43.3	26.4	26.0	300	0.765

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.7	70.5	64.8	68.1	63.5	66.0	62.3	66.3	69.0	64.5	66.9	63.0	65.1	10.3	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.2	93.5	67.6	63.5	87.9	65.8	62.0	83.3	64.3	31.0	69.0	29.7	67.0	28.6	65.2	74.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
40.8	36.1	31.9	DB	6.0	76.2	4.3	2.9	2.9	78.3	0.4	80.0	-2.0	81.7	-5.1	83.8
			WB	5.5	69.9	3.9	2.1	2.7	71.5	0.4	72.7	-1.8	73.9	-4.6	75.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	45.9	29.8	30.1	34.4	41.6	48.4	54.6	59.9	60.9	58.0	52.0	43.9	35.7	(6)
(7)		DBStd	12.32	8.64	7.54	6.30	4.40	3.75	3.82	3.25	2.91	3.64	4.83	6.63	7.55	(7)
(8)		HDD50	2682	626	557	483	253	77	6	0	0	0	36	200	444	(8)
(9)		HDD65	6989	1091	977	947	702	514	311	161	131	211	403	633	908	(9)
(10)		CDD50	1176	0	0	0	1	28	145	307	338	241	98	17	1	(10)
(11)		CDD65	8	0	0	0	0	0	0	3	3	2	0	0	0	(11)
(12)		CDH74	11	0	0	0	0	0	0	5	3	3	0	0	0	(12)
(13)		CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	14.7	19.8	19.3	17.6	14.7	11.8	10.8	10.0	9.4	10.7	15.1	17.9	19.5	(14)
(15)	Precipitation	PrecAvg	49.6	4.4	3.8	4.2	3.7	3.7	3.4	3.3	3.1	4.2	4.8	4.9	5.4	(15)
(16)		PrecMax	66.7	8.3	7.6	8.5	7.0	8.0	10.4	6.6	7.2	10.3	11.7	8.7	10.4	(16)
(17)		PrecMin	33.0	1.7	1.5	0.7	0.5	1.1	1.2	0.9	1.1	2.2	1.6	1.9	2.1	(17)
(18)		PrecStd	7.2	1.4	1.4	1.7	1.6	1.8	2.0	1.7	1.5	2.1	2.2	1.7	1.9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.3	48.1	52.0	55.7	63.8	69.5	74.0	73.6	71.6	65.8	61.1	54.9	(19)
(20)			MCWB	50.0	46.7	48.3	50.5	55.6	61.9	66.7	66.7	66.2	62.8	59.0	53.8	(20)
(21)		2%	DB	47.8	44.6	47.5	51.9	59.0	65.4	70.1	70.3	68.0	62.7	57.8	51.9	(21)
(22)			MCWB	46.7	43.1	45.0	47.6	53.1	59.8	64.2	65.2	64.1	60.3	56.2	51.1	(22)
(23)		5%	DB	44.9	42.2	44.8	49.4	56.3	62.7	67.4	68.0	65.2	60.5	55.3	49.0	(23)
(24)			MCWB	43.4	40.6	42.6	46.1	52.0	58.2	62.9	64.0	62.2	58.0	53.5	47.5	(24)
(25)		10%	DB	41.9	40.2	42.2	47.4	54.0	60.4	65.2	65.9	63.3	58.7	53.0	46.2	(25)
(26)			MCWB	40.0	38.4	40.1	44.5	50.4	56.5	61.7	62.6	60.8	56.5	51.0	44.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.5	47.1	49.5	53.4	58.0	64.0	68.3	69.1	67.5	63.6	59.8	54.1	(27)
(28)			MCD B	51.1	47.7	51.3	54.5	61.4	67.5	71.6	71.5	69.8	64.9	60.7	54.6	(28)
(29)		2%	WB	46.8	43.7	45.6	49.1	54.9	61.3	65.6	66.5	64.8	60.8	56.6	51.2	(29)
(30)			MCD B	47.6	44.5	47.0	50.6	57.5	64.2	68.8	69.0	66.8	62.0	57.6	51.8	(30)
(31)		5%	WB	43.8	41.1	43.0	46.8	52.7	59.1	63.8	64.7	63.0	58.7	53.9	47.9	(31)
(32)			MCD B	44.7	42.0	44.6	48.6	55.3	61.6	66.3	67.0	64.6	60.1	55.0	48.7	(32)
(33)		10%	WB	40.2	38.7	40.5	45.0	50.9	57.2	62.2	63.1	61.2	56.9	51.3	44.7	(33)
(34)			MCD B	41.6	39.9	41.8	46.9	53.3	59.4	64.5	65.2	62.9	58.4	52.7	46.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.3	9.7	8.6	8.4	9.6	10.9	11.3	10.7	8.8	7.5	8.1	9.2	(35)
(36)			MCDBR	14.9	11.7	11.8	11.3	13.4	14.6	14.7	13.6	11.7	9.7	10.3	12.7	(36)
(37)			MCWBR	15.7	11.8	10.6	8.2	8.8	9.7	9.8	9.4	8.6	8.3	10.7	13.6	(37)
(38)		5% WB	MCDBR	15.0	11.9	12.3	10.7	12.3	13.5	13.4	12.4	10.9	9.3	10.7	12.6	(38)
(39)			MCWBR	16.0	12.5	11.6	9.0	9.3	10.3	10.2	9.8	9.1	9.0	11.5	13.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.306	0.308	0.333	0.355	0.378	0.408	0.424	0.405	0.368	0.341	0.324	0.308	(40)	
(41)		taud	2.479	2.498	2.454	2.429	2.411	2.342	2.327	2.379	2.444	2.518	2.523	2.491	(41)	
(42)		Ebn at Noon	260	282	287	288	283	274	268	269	272	266	252	248	(42)	
(43)		Edh at Noon	22	26	31	35	36	39	39	36	31	25	22	20	(43)	
(44)	All-Sky Solar Radiation	RadAvg	407	698	1111	1422	1615	1785	1794	1617	1326	854	481	334	(44)	
(45)		RadStd	36	62	84	137	200	170	142	100	91	90	53	33	(45)	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	+0.88	N/A	N/A	N/A	N/A	+124	+2